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Intermolecular decoupling of C_{60} suppresses interfacial recombination in perovskite solar cells

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Interfacial non-radiative recombination, particularly at the C_{60} electron-selective contact, remains a major bottleneck limiting the performance of inverted perovskite solar cells. While previous efforts have primarily focused on passivating trap states at the perovskite surface, the underlying mechanism by which fullerene-based electron transport layers induce interfacial recombination has remained unclear. Here we reveal that C_{60} lowest unoccupied molecular orbital (LUMO)-derived band-like states arising from strong intermolecular coupling within the C_{60} film drive recombination at the perovskite/C_{60} interface. To mitigate this issue, we develop a vapour-phase molecular passivation strategy using trialkylamines to disrupt C_{60} molecular packing, substantially weakening the C_{60} LUMO-derived band-like states and suppressing interfacial non-radiative losses. The trialkylamine treatment minimizes the quasi-Fermi-level splitting loss after C_{60} deposition from 89 meV to 41 meV, and to 17 meV in conjunction with standard surface passivation. This interfacial modulation enables inverted perovskite solar cells with a certified power conversion efficiency of 27.2% (stabilized 26.6%) on a 1.02 cm^2 active area. The devices also retain over 98% of their initial efficiency after 1,000 h of continuous illumination under ambient air conditions. Our work resolves the long-standing puzzle of why C_{60} induces severe interfacial recombination and guides the design of comprehensive interface passivation strategies towards efficient perovskite photovoltaics.

The certified power conversion efficiency (PCE) of inverted p–i–n perovskite solar cells (PSCs) has surpassed 27.0%, outperforming regular n–i–p counterparts[1,2]. To further approach the Shockley–Queisser limit, non-radiative recombination at the perovskite/carrier transport layer interfaces must be systematically minimized[3,4]. The recombination at the perovskite/hole transport layer interface has been addressed by self-assembled monolayers[5–7]. However, the perovskite/electron transport layer interface, typically involving C_{60}, still suffers from substantial non-radiative recombination losses, remaining a key bottleneck for advancing single- and multi-junction perovskite photovoltaics and industrial modules[8–10].

Previous efforts targeting the perovskite/C_{60} interface have largely focused on chemical passivation strategies that reduce surface defect density and enhance the photoluminescence quantum yield (PLQY)[11]. However, such improvements often diminish after C_{60} deposition[9]. Recent studies attribute the C_{60}-induced PLQY decline to the across-interface recombination, where electrons extracted into C_{60} are trapped and recombine with holes in the perovskite, rather than the intrinsic perovskite defects[9,12]. Field-effect passivation strategies, including dipolar compounds[8,9] and metal fluoride interlayers[10,13], mitigate this recombination primarily by repelling near-interface holes and blocking recombination channels. Despite these advances, why C_{60}

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introduces substantial non-radiative recombination across the perovskite/C_{60} interface remains unclear, and state-of-the-art inverted PSCs, especially large-scale devices, still exhibit large PCE deficits compared with the radiative limit, underscoring the need for deeper mechanistic understanding and more comprehensive suppression strategies.

Here, we identified C_{60} lowest unoccupied molecular orbital (LUMO)-derived band-like states as the primary origin of fast non-radiative recombination at the perovskite/C_{60} interface through theoretical calculations. We developed a vapour-phase molecular passivation strategy that disrupts C_{60} packing and suppresses C_{60}-induced interfacial recombination, enabling PSCs with a certified PCE exceeding 27.0% and good operational stability.

Theoretical investigations

To investigate the origin of C_{60}-induced recombination loss, we performed density functional theory (DFT) calculations on isolated C_{60} atop a prototypical $FAPbI_3$ perovskite framework (Fig. 1a), following previous studies[12–14]. The calculated density of states (DOS) revealed no gap states within the perovskite bandgap (Fig. 1b), consistent with the previous calculation[12], regardless of whether the C_{60} molecule sat atop a Pb or an I atom (Supplementary Fig. 1). Also, no gap states were observed for the isolated C_{60} molecule in vacuum without the perovskite framework (Supplementary Fig. 2).

We hypothesized that the isolated C_{60} model overlooked the intermolecular interactions present in deposited C_{60} films. To account for these effects, we modelled small C_{60} clusters on the perovskite surface (Supplementary Fig. 3). The calculated DOS still revealed no gap states for either the C_{60} dimer or trimer clusters, suggesting that short-range interactions were insufficient to induce electronic defects. This changed when periodic molecular ordering was introduced. We constructed a single-row C_{60} nanowire under one-dimension periodic conditions (Supplementary Fig. 4a,b), where gap states began to emerge. These states became more pronounced in the double-row nanowire (Supplementary Fig. 4c,d), indicating that periodic ordering drives the formation of band-like states.

Then, we modelled a monolayer C_{60} configuration on the perovskite framework and optimized the lowest-energy interfacial distance (Fig. 1c and Supplementary Figs. 5 and 6). The resulting DOS showed distinct mid-gap states (Fig. 1d), contributed almost entirely by C_{60} carbon atoms. This explains the experimentally observed PLQY quenching after deposition of the first C_{60} layer[12]. A bilayer C_{60} configuration yielded similar results, with more broadened mid-gap states (Supplementary Fig. 7).

To decouple interfacial effects, we artificially increased the perovskite–C_{60} separation to 1 Å, 2 Å, 3 Å (Supplementary Fig. 8) and 5 Å (Fig. 1e). Although the DOS slightly decreased with increasing separation, prominent mid-gap states persisted (Fig. 1f), suggesting that the perovskite–C_{60} interaction contributed only minimally to their formation. Similar band-like states appeared in bulk C_{60} models (Supplementary Fig. 9), consistent with previous computational results[15,16], and are corroborated by the photoemission and inverse-photoemission studies confirming delocalized states and band dispersion in solid C_{60} (ref. 17).

Together, these findings indicate that the band-like states originate from the long-range C_{60} molecular ordering, which enables π–π intermolecular coupling and leads to the formation of band-like states derived from the periodic C_{60} LUMOs via extended orbital overlap[18]. We believe that these quasi-delocalized states are insufficiently localized to act as direct recombination centres, as they efficiently trap electrons but intrinsically capture holes poorly. Instead, they accelerate recombination via existing defects at the perovskite/C_{60} interface (Fig. 1g). To quantify this, we modelled the electron capture coefficient (C) from the perovskite conduction band (CB) through the LUMO of isolated or periodic C_{60} into existing perovskite surface defects (Fig. 1h), following Kirchartz and Markvart[19,20], as a function of energy difference ΔE between C_{60}-LUMO and defect level (Methods). The increasing C as ΔE decreases is known as 'energy-gap-law'[21]. The recombination rate (R) then increases linearly with C, given by $R = CnN_T$, with n being the carrier density and N_T the number of existing surface defects. Interestingly, periodic C_{60}, which forms a lower-lying 'LUMO-band' between the perovskite CB and a deep defect, allows a cascade-like recombination and leads to high capture coefficients. Disrupting the C_{60} LUMO-derived band-like states can lower the capture coefficient into both neutral and positively charged defects dramatically (by three orders of magnitude). This allows a drastic reduction of surface recombination, even without (chemically) passivating existing surface defects. Furthermore, as shown in Supplementary Fig. 3e, the periodic extension of the primitive unit cell from Supplementary Fig. 3c yielded spatially separated C_{60} trimer clusters that resemble a periodicity-broken single-row nanowire and did not induce band-like states.

These insights motivated us to minimize the band-like states by decoupling C_{60} packing. Accordingly, we developed a vapour-phase passivation strategy using trialkylamines by exposing C_{60}-coated samples to trialkylamine vapour under vacuum (Supplementary Fig. 11). Our motivation to use this trialkylamine was guided by their interactions with C_{60} (Supplementary Figs. 12–14) and experimental screening of different amines (Supplementary Fig. 15).

We calculated six interfacial configurations for monolayer C_{60} with inserted trihexylamine (C18N) (Supplementary Fig. 16). The most energetically favourable configuration (Fig. 1i) exhibited significantly reduced C_{60} LUMO-derived band-like states (Fig. 1j). A similar decline was observed for bilayer C_{60} (Supplementary Figs. 17 and 18). The DFT-predicted reduction in mid-gap states was supported by the reduced sub-bandgap external quantum efficiency (EQE) response (Supplementary Fig. 19). The DFT results with tributylamine (C12N) and trioctylamine (C24N) insertion (Supplementary Fig. 20) further revealed chain-length dependence, where C12N (short-chain) is insufficient to decouple intermolecular C_{60} interactions, while C24N (long-chain) prefers a flat-lying interfacial configuration rather than insertion into the C_{60} matrix. Both cases showed more limited reduction in mid-gap states.

Morphological characteristics

To examine C18N distribution in PSCs, we conducted time-of-flight secondary ion mass spectrometry (TOF-SIMS). The $C_6H_{14}^-$ fragment, assigned to C18N, was absent in the pristine device (Supplementary Fig. 21), while it was enriched at both the bottom and top C_{60} interfaces in the C18N-treated device (Fig. 2a). The $C_6H_{14}^-$ signal decreased in parallel with the C_2H^- signal from C_{60}, indicating that C18N molecules were primarily confined to the C_{60} layer rather than penetrating the bulk perovskite. This interfacial enrichment made it possible to disrupt C_{60} packing and modulate interfacial band-like states. Molecular dynamics (MD) simulations further showed that C18N progressively accumulated near the perovskite/C_{60} interface on the C_{60} side (Supplementary Fig. 22), driven by weak miscibility with closely packed C_{60} and the resulting excluded-volume effect.

Two-dimensional grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were performed to investigate morphological changes induced by the trialkylamine treatment (Fig. 2b,c). Both the pristine and C18N-treated C_{60} films exhibited halo-like diffraction patterns, with no additional diffraction rings observed, suggesting preserved randomly oriented crystallites and no phase transformation due to C18N incorporation. However, the apparent decline in diffraction intensity indicated reduced crystallinity and disrupted packing symmetry in the C18N-treated C_{60} film, consistent with the structural scenario depicted in Fig. 1i and Supplementary Fig. 17, while pole figure analysis of the (220) plane attributes the intensity loss primarily to amorphization rather than crystallite reorientation (Supplementary Fig. 23). One-dimensional GIWAXS analysis revealed decreased scattering vector q values of the (111), (220) and (311) planes[22], corresponding to

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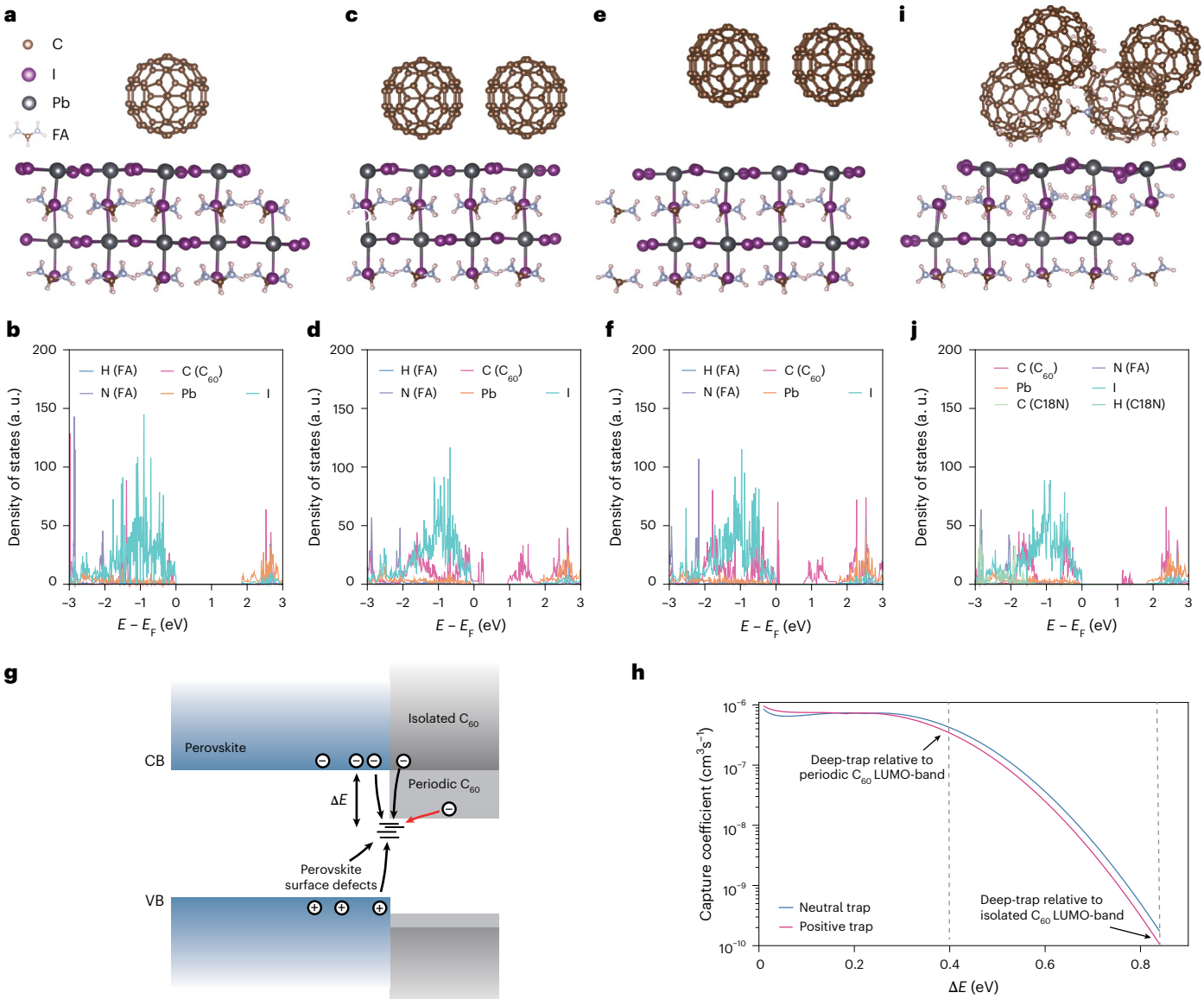


Fig. 1 | Computational modelling of the interface between C_{60} and the perovskite framework. a–**f**, **i** and **j**, The sideview of interface configurations modelled by DFT (**a**, **c**, **e** and **i**) and DOS contributions from different elements (**b**, **d**, **f** and **j**) for isolated C_{60} positioned atop a Pb atom on the perovskite framework (**a** and **b**), monolayer C_{60} on the perovskite framework with optimized distance from the perovskite (**c** and **d**), monolayer C_{60} on the perovskite framework with a fixed 5-Å separation between the perovskite and C_{60} (**e** and **f**), and monolayer C_{60} on the perovskite framework with C18N insertion, the most energetically favourable configuration (**i** and **j**). **g**, An energetic scheme of the interface, including possible defect levels. VB, valence band. **h**, The capture coefficient for neutral defects, or with a trapped hole (pos.defect), as a function of energy difference between the between C_{60}-LUMO and defect level with exemplary indications for electron recombination from C_{60} to perovskite surface defects. The coefficient could increase by several magnitudes if C_{60} occurs as packed/crystalline structure forming a 'LUMO-band'.

increased lattice *d*-spacing values ($d = 2\pi/q$) of C_{60} (Fig. 2d), in agreement with the lattice expansion observed in the computational results above. In particular, the average (111) spacing increased by 0.16 Å after treatment. As C18N was distributed at both C_{60} interfaces, the lattice expansion is expected to occur at the bottom interface as well. Atomic force microscopy (AFM) images (Supplementary Fig. 24) showed no significant topographic variation in C_{60} films on indium tin oxide (ITO).

To validate elemental distribution and the morphology, we performed high-resolution transmission electron microscopy (HR-TEM) with electron energy loss spectroscopy (EELS) on cross-sectional samples. The ITO/evaporated PbI_2/C_{60}/SnO_x/Ag stack was employed to eliminate interference from nitrogen-containing components in the perovskite and solvents, ensuring detected nitrogen signals originated solely from C18N. Elemental mapping by energy-dispersive X-ray spectroscopy (EDX) (Fig. 2e) showed tin, carbon and lead distributions matching the atomic layer deposition (ALD)-SnO_x, C_{60} and PbI_2 layers, respectively. EELS spectra (Fig. 2f) showed nitrogen signals localized at the top and bottom interfaces of the C18N-treated C_{60} layer, but more pronounced at the bottom interface, consistent with the TOF-SIMS analysis, whereas negligible nitrogen signal was observed in the pristine stack (Supplementary Fig. 25). Selected area electron diffraction (SAED) at the bottom C_{60} interface in the ITO/perovskite/C_{60}/SnO_x/Ag stacks exhibited diffuse diffraction spots and amorphous rings (Fig. 2g,h). The attenuation of diffraction spots after C18N treatment indicated disrupted molecular periodicity. Cross-sectional HR-TEM imaging (Supplementary Fig. 26) further revealed the coexistence of crystalline and amorphous domains, confirming the polycrystalline nature of the evaporated C_{60} films. The (111) diffraction ring contracted

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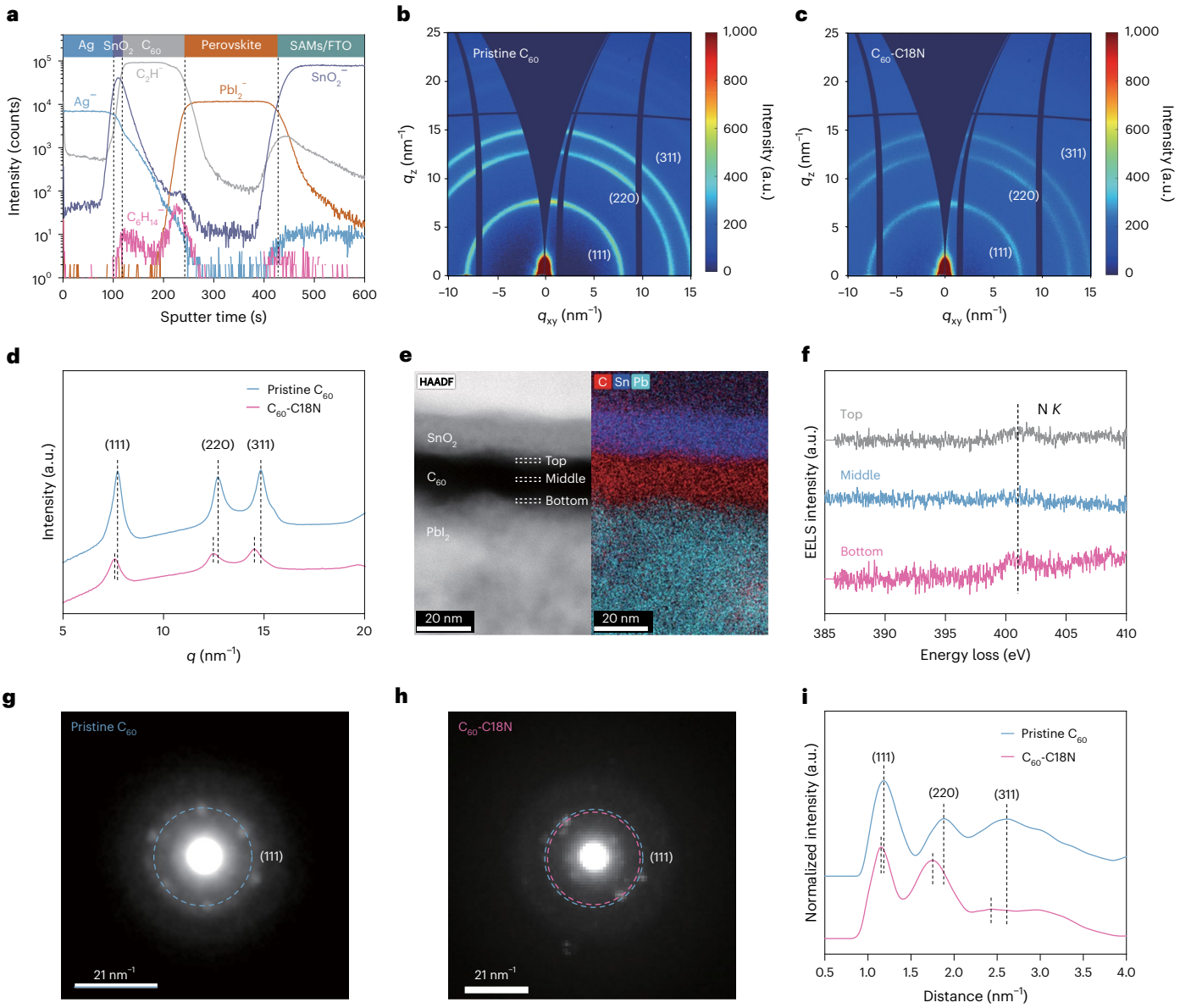


Fig. 2 | Distribution of C18N and crystallinity changes in C_{60} films after C18N treatment. a, TOF-SIMS depth profiles of the PSCs with C18N-treated C_{60} film. **b**,**c**, GIWAXS patterns of pristine (**b**) and C18N-treated (**c**) C_{60} films on Si substrates. **d**, One-dimensional GIWAXS scattering profiles of the pristine and C18N-treated C_{60} films. **e**, A high-angle annular dark-field (HAADF) image from HR-TEM and corresponding EDX elemental mapping of the ITO/PbI_2/C_{60}/SnO_x/Ag stack. **f**, EELS spectra from three regions marked in **e** of C18N-treated stacks. **g**,**h**, SAED patterns collected from the bottom interface of the pristine (**g**) and C18N-treated (**h**) C_{60} layers in the ITO/perovskite/C_{60}/SnO_2/Ag stacks. The blue and pink circles represent the (111) plane of the pristine and C18N-treated C_{60} films, respectively. **i**, Integral intensity profiles of SAED for the pristine and C18N-treated stacks.

following C18N incorporation, further evidencing lattice expansion (Fig. 2i). Integral intensity profiles of the SAED patterns revealed the diffraction ring narrowing, with the (111) plane expanding by an average of 0.25 Å (d = 1/distance from the centre). The expansion was more pronounced at the bottom interface than the top interface (Fig. 2d), probably reflecting the higher C18N concentration there (Fig. 2a).

Passivation effects

To evaluate whether the disruption of C_{60} packing contributes to interfacial passivation, as predicted computationally, we quantified the non-radiative recombination losses at the perovskite/C_{60} interface using PLQY measurements (Fig. 3a). Without the C_{60} layer, PLQY slightly decreased after both 1,3-propanediammonium iodide ($PDAI_2$) deposition and subsequent C18N treatment, indicating neither treatment chemically passivated the perovskite surface[8]. After C_{60} deposition, C18N treatment markedly increased PLQY, even on top of the $PDAI_2$-treated sample that already exhibited improved PLQY. The C18N treatment reduced the quasi-Fermi-level splitting (QFLS) loss after C_{60} deposition from 89 to 41 meV, and further to 17 meV when combined with $PDAI_2$ (Supplementary Fig. 27). Although $PDAI_2$ effectively blocks holes from the perovskite side, with limited influence on C_{60} intermolecular coupling (Supplementary Fig. 28), C18N further suppresses low-lying states near C_{60} sides and minimizes the non-radiative recombination across the perovskite/C_{60} interface.

We also quantified light intensity-dependent QFLS, and plotted implied V_{OC} (iV_{OC}) against illumination intensity (Fig. 3b) and constructed pseudo-current density–voltage (pseudo-J–V) curves (Fig. 3c) for pristine and C18N-treated C_{60} stacks. $PDAI_2$-treated perovskite on ITO/self-assembled monolayers (SAMs) substrates exhibited impressive pseudo photovoltaic performance. However, following

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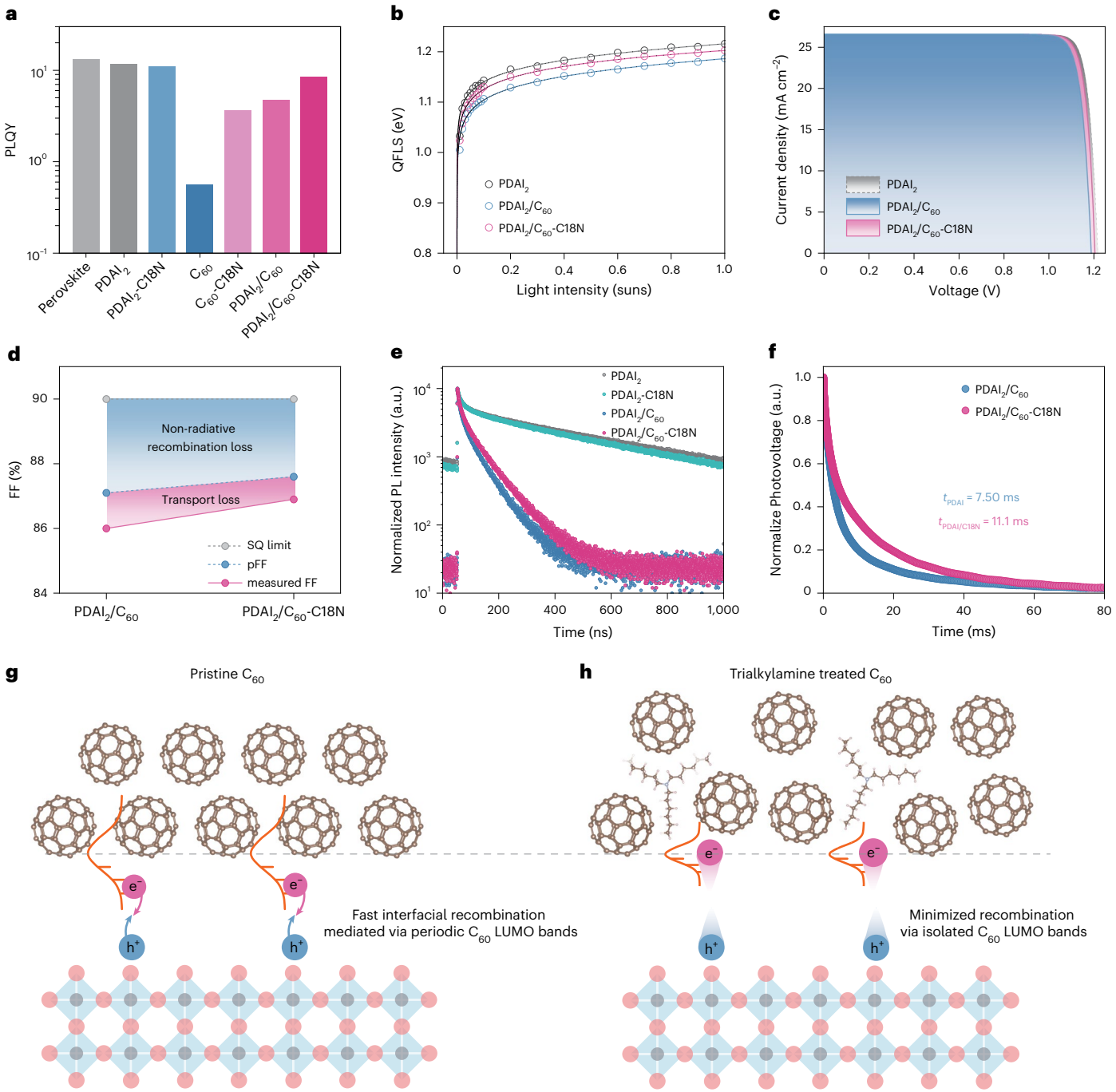


Fig. 3 | Passivation effect of C18N treatment. a, PLQY data of the perovskite films without and with a C_{60} layer, before and after C18N treatment on ITO/SAMs substrates. **b**, Light intensity-dependent QFLS plots of $PDAI_2$ passivated perovskite films without and with pristine and C18N-treated C_{60} layers. The fitting curves are obtained through exponential fitting. **c**, Pseudo-*J*–*V* curves derived from the light intensity-dependent QFLS data. **d**, FF loss analysis for $PDAI_2$- and $PDAI_2$/C18N-treated devices. SQ, Shockley–Queisser. **e**, TRPL decay profiles of $PDAI_2$ passivated perovskite films without and with C_{60} layers, before and after C18N treatment. **f**, Transient photovoltage plots for $PDAI_2$- and $PDAI_2$/C18N-treated PSCs. **g**,**h**, Schematic diagram showing interfacial recombination for pristine C_{60} (**g**) and its suppression by trialkylamine treatment (**h**).

C_{60} deposition, inadequately suppressing interfacial band-like states led to a reduction in the implied device performance. This degradation was mitigated in the C18N-treated C_{60} stack, which achieved a pseudo-PCE of 27.9%, with an iV_{OC} of 1.203 V and pseudo-fill factor (FF) of 87.6%. These values exceed those of the pristine C_{60} stack, which showed a pseudo-PCE of 27.4%, iV_{OC} of 1.187 V and pseudo-FF of 87.1%. Pseudo-FF and actual device FF are compared in Fig. 3d. The FF loss caused by trap-assisted non-radiative recombination was also reduced, yielding a 0.5% FF gain in the C18N-treated device.

Time-resolved photoluminescence (TRPL) measurements revealed a slight decrease in photoluminescence (PL) lifetime for the perovskite alone and an increase for the perovskite/C_{60} stack after C18N treatment (Fig. 3e and Supplementary Table 2), confirming that C18N primarily suppressed non-radiative recombination at the C_{60} interfaces. To separate charge extraction from trap-assisted recombination, we analysed the differential lifetime, defined as $\tau = -\{d\ln[\phi(t)]/dt\}^{-1}$, where $\phi(t)$ is the time-dependent PL photon flux[5] (Supplementary Fig. 29). The steeper initial rise of τ (until ~200 ns) indicated faster electron transfer

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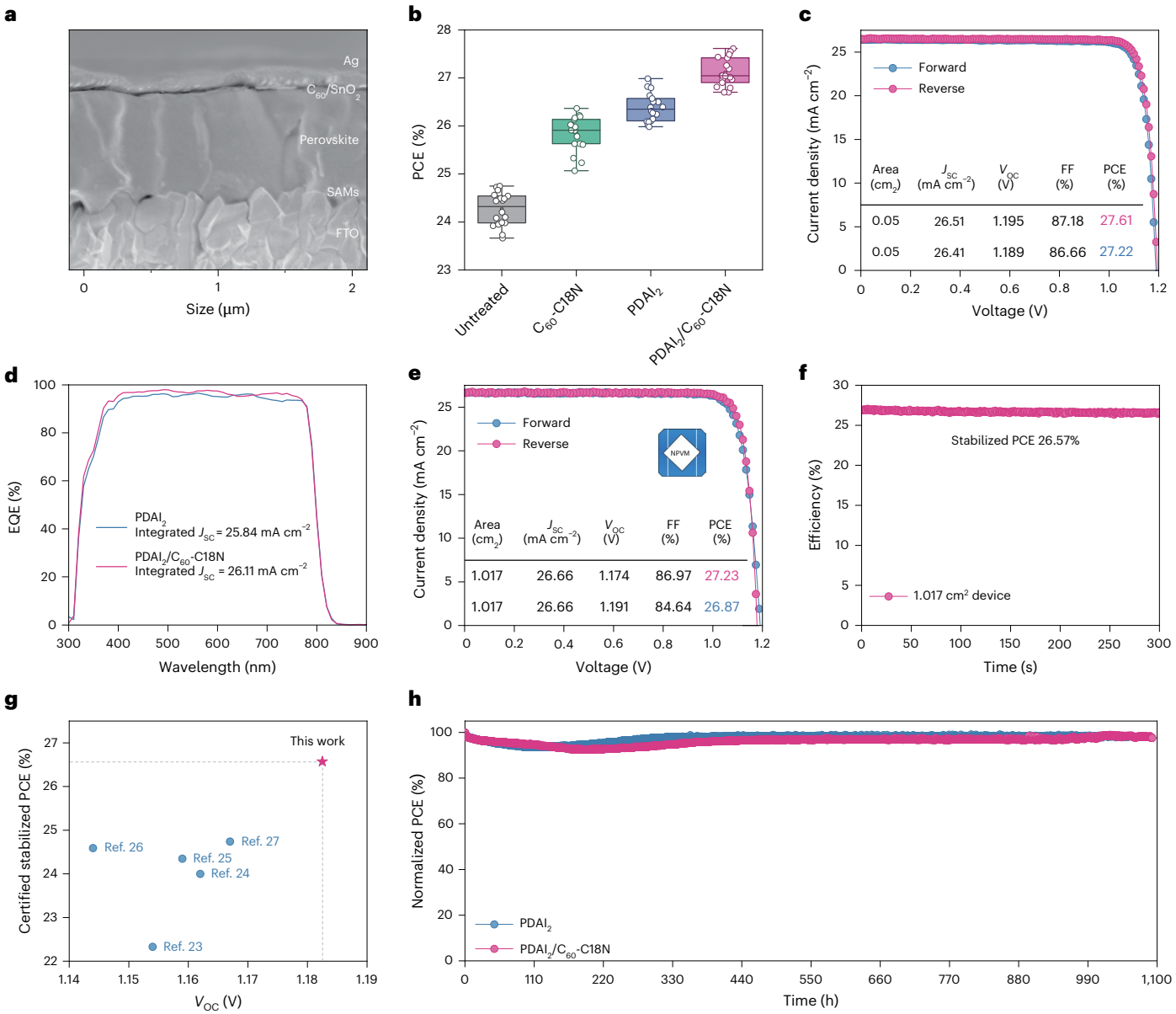


Fig. 4 | Photovoltaic performance and long-term stability of the PSCs. **a**, A cross-sectional SEM image of the PSCs. **b**, PCE statistics for unpassivated, C18N-treated C_{60}-based, $PDAI_2$-treated and $PDAI_2$/C18N-treated PSCs (18 devices for each type). The centre line represents the median, the box limits indicate upper and lower quartiles, and the whiskers denote the minimum and maximum values. **c**, *J*–*V* curves of the best 0.05 cm^2 C18N-treated device. **d**, An EQE comparison between the control and C18N-treated PSCs. **e**,**f**, A *J*–*V* scan (**e**) and MPPT measurements (**f**) of the 1.02 cm^2 device recorded by NPVM. **g**, Certified stabilized PCE statistics for reported state-of-the-art 1 cm^2 PSCs[23–27]. **h**, Long-term operational stability of encapsulated devices under 1-sun illumination (100 mW cm^{-2}) at room temperature in air with 60% RH.

from perovskite to C_{60}. Transient photovoltage (Fig. 3f) and photocurrent (Supplementary Fig. 30) measurements of full device stacks further supported this finding, revealing suppressed recombination and more efficient electron extraction via slower photovoltage decay and faster photocurrent decay. C18N-treated PSCs exhibited lower dark current (Supplementary Fig. 31), attributed to the suppressed interfacial carrier recombination and increased charge collection. Mott–Schottky analysis revealed that the built-in potential (V_{bi}) of PSCs increased from 0.99 to 1.01 V after C18N treatment (Supplementary Fig. 32), indicating an enhanced internal electric field for carrier separation at the C_{60} interface. These observations were consistent with our computationally predicted reduction in C_{60} LUMO-derived band-like states.

To investigate the interface electrical-field distribution after C18N treatment, we conducted cross-sectional Kelvin probe force microscopy (KPFM) measurements on full devices (Supplementary Fig. 33). Distinct electrical-field spikes were observed at both perovskite interfaces, indicating regions of charge separation. Upon introducing C18N on the $PDAI_2$-based device, the interfacial electric field, particularly at the perovskite/C_{60} interface, showed significant enhancement, suggesting improved charge separation, probably from the enhanced internal electric field and the retained electron conductivity (Supplementary Fig. 34). Ultrafast transient absorption (fs-TA) probed charge dynamics. The pseudocolor fs-TA maps (Supplementary Fig. 35) displayed a positive photoabsorption signal between 650 and 700 nm and a negative photobleaching peak around 720–800 nm, with reduced photobleaching peak intensity for the perovskite/C_{60}-C18N sample, indicating faster carrier transfer to C_{60}. Decay kinetics at 767 nm were well fitted using a tri-exponential decay model (Supplementary Fig. 35e and Supplementary Table 3), with τ_1 assigned to hot-carrier extraction, τ_2 to charge-carrier trapping and τ_3 to exciton recombination.

The C18N-treated sample exhibited a shorter τ_1, indicative of enhanced electron extraction, and increased τ_2 and τ_3, suggesting suppressed non-radiative recombination, consistent with TRPL results.

In sum, C18N treatment disrupts C_{60} packing symmetry, suppresses the C_{60} LUMO-derived band-like states and minimizes C_{60}-induced fast interfacial non-radiative recombination, as depicted in Fig. 3g,h.

Photovoltaic performance and stability

To evaluate the passivation effect of C_{60} on device efficiency and stability, we fabricated inverted PSCs with the structure fluorine-doped tin oxide (FTO)/(4-(2,7-dibromo-9,9-dimethylacridin-10(9H)-yl) butyl)phosphonic acid/perovskite/C_{60}/SnO_x/Ag (Fig. 4a). Vacuuming time was optimized to 6 min (Supplementary Fig. 36), and among the trialkylamines tested, C18N gave the best device performance (Supplementary Fig. 37). Device statistics (Fig. 4b) revealed substantial PCE gains, reaching 25.8% for C18N-treated PSCs and 26.4% for the $PDAI_2$-treated PSCs, compared with 24.3% for unpassivated devices[8]. Applying C18N treatment to $PDAI_2$-treated PSCs further increased the average PCE to 27.1% without mutual interference, as indicated by consistent improvements across all photovoltaic parameters (Supplementary Fig. 38), attributable to minimized interfacial recombination. Devices with an active area of 0.05 cm^2 achieved a maximum PCE of 27.6%, with a V_{OC} of 1.195 V, a short-circuit current density (J_{SC}) of 26.5 mA cm^{-2} and an FF of 87.1% (Fig. 4c) under reverse scan. C18N treatment on $PDAI_2$-treated perovskite before C_{60} evaporation reduced the PCE, mainly owing to a lower FF (Supplementary Fig. 40), suggesting surface-adsorbed C18N formed an insulating layer that increased series resistance rather than passivating the interface. Spin-coated C18N on C_{60} similarly gave inferior performance, accompanied by a more uniform bulk distribution in the C_{60} layer (Supplementary Fig. 41). These results highlight that vapour-phase treatment is essential for effective interfacial passivation.

Integrated J_{SC} increased from 25.8 to 26.1 mA cm^{-2} after C18N treatment (Fig. 4d), primarily from suppressed electron trapping and improved extraction. The reduced slope of the light intensity-dependence V_{OC} plot for C18N-treated devices (Supplementary Fig. 42) indicated suppressed low-lying states at the perovskite/C_{60} interface. Electrochemical impedance spectroscopy (EIS) measurements revealed lower series resistance (R_s) and a higher recombination resistance (R_{rec}) for C18N-treated devices (Supplementary Fig. 43 and Supplementary Table 4), suggesting more efficient transport and suppressed surface recombination, both contributing to the enhanced V_{OC} and FF.

The large-area device (1.02 cm^2) achieved a National PV Industry Measurement and Testing Center (NPVM)-certified PCE of 27.2% (Fig. 4e) and a stabilized PCE of 26.6% under maximum power point tracking (MPPT) over 300 s (Fig. 4f and Supplementary Fig. 44). The certified stabilized PCE represents the highest efficiency reported so far for 1 cm^2 PSCs[23–27] (Fig. 4g and Supplementary Table 5). Similar improvements were observed in 1.68 eV wide-bandgap perovskite devices (Supplementary Fig. 45). The passivation strategy minimally affected operational or thermal stability. Encapsulated $PDAI_2$/C18N and $PDAI_2$-based devices both retained over 98% of initial performance after operating under simulated AM1.5G radiation (100 mW cm^{-2}) in ambient air (~60% relative humidity (RH)) for 1,000 h (Fig. 4h). The 'dip–recover–decay' behaviour reflects the transition from reversible light-soaking effects to irreversible degradation processes[28,29]. C18N was retained at the interface without noticeable loss or redistribution during operation (Supplementary Fig. 46). Non-encapsulated devices annealed at 85 °C in nitrogen maintained ~70% of initial PCE after 1,348 h under both conditions (Supplementary Fig. 47). The limited stability improvement may result from insufficient suppression of ion migration, which dominates the degradation process, while C18N primarily mitigates interfacial deterioration rather than intrinsic perovskite instability[30,31].

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41566-026-02030-x.

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Methods

Materials

All chemicals were used as received. Formamidinium iodide, lead iodide (PbI_2, 99.99%) and all the trialkylamine molecules were purchased from Tokyo Chemical Industry. Methanamine hydriodide and methylamine hydrochloride were purchased from Greatcell Solar Materials. (4-(2,7-Dibromo-9,9-dimethylacridin-10(9H)-yl)butyl) phosphonic acid and $PDAI_2$ were purchased from Vizuchem. Fullerene (C_{60}, >99.5%) was purchased from Luminescence Technology. Caesium iodide (CsI, 99.99%) and all anhydrous solvents were purchased from Sigma-Aldrich.

Processing methodology

The C_{60}-covered samples were removed from the glovebox and placed in an custom-built acrylic chamber (30 × 16 × 16 cm^3) along with a 4-ml open vial containing 20 µl of trialkylamine liquid. The chamber was then sealed and evacuated to −0.1 MPa for 6 min, and the deposition level could be tuned by the processing duration. After the chemical treatment, the samples were returned to the glovebox and annealed at 75 °C for 5 min to remove any residual trialkylamine. The room temperature and relative humidity in our perovskite laboratory in Singapore were maintained at approximately 18 °C and 70% RH. If the ambient humidity is very low, a similar preconditioning step using deionized water before the trialkylamine treatment should be performed to recover optimal interfacial passivation (Supplementary Fig. 39). The exposure time may require minor re-optimization in different laboratory setups.

Solar cell fabrication

FTO glasses were cleaned sequentially with soap, deionized water, acetone and isopropanol under ultrasonication, and then dried with dry nitrogen. Before use, the substrates were treated with ultraviolet ozone for 15 min and transferred to a nitrogen glovebox. (4-(2,7-Dibromo-9,9-dimethylacridin-10(9H)-yl)butyl)phosphonic acid (0.5 mg ml^{-1} in ethanol) was spin-coated on the FTO substrates at 4,000 rpm for 30 s and annealed at 100 °C for 10 min (ref. 6). The 1.55 M $Cs_{0.05}FA_{0.85}MA_{0.1}PbI_3$ perovskite precursor solution was prepared by mixing 0.0775 mmol CsI, 1.32 mmol formamidinium iodide, 0.155 mmol methanamine hydriodide, 1.69 mmol PbI_2 (9 mol% excess) and 0.232 mmol methylamine hydrochloride in 1 ml of mixed DMF:DMSO (4:1, v:v) solvent. The precursor solution was shaken overnight and then filtered through a 0.22-µm PTFE membrane before use. Then, 100 µl of perovskite solutions were spin-coated onto the glass/FTO/SAMs substrates at 2,000 rpm for 20 s and 5,000 rpm for 35 s. Then, 120 µl of anisole was dripped onto the centre of the spinning substrates at 10 s before the end of spin-coating. The as-prepared perovskite films were subsequently annealed on a hotplate at 100 °C for 15 min. After cooling to room temperature, 100 µl of 0.5 mg ml^{-1} $PDAI_2$ in isopropanol was spin-coated on the perovskite surface at 5,000 rpm for 30 s and annealed at 100 °C for 5 min. Following this, 20 nm C_{60} was subsequently deposited on top by thermal evaporation at a rate of 0.2 Å s^{-1} under a vacuum (<10^{-5} Pa). After chemical treatment, the substrates were transferred to an ALD system (Ke-micro) to deposit 20 nm SnO_x at 100 °C using precursors of tetrakis(dimethylamino) tin(IV) (99.9999%) and deionized water. Finally, a 100 nm Ag electrode was deposited by thermal evaporation.

Computational details

All spin-polarized DFT calculations were performed using the Vienna Ab initio Simulation Package (version 5.4.4)[32] under the framework of the projector augmented wave method[32,33]. The generalized gradient approximation of the Perdew–Burke–Ernzerhof functional was employed as the exchange-correlation functional[34]. The PBE+D3 method for the van der Waals correction was also included[35]. The valence electronic wave functions were expanded in plane waves with a cutoff energy of 400 eV, and the convergence criteria of force and energy were set to 0.02 eV $Å^{-1}$ and 10^{-4} eV, respectively. The 2 × 2 (001) surface of perovskites ($FAPbI_3$) with PbI_2 termination was constructed as the surface model. C_{60} with one layer or two stacked layers were further added to the perovskite surface to construct the interface models. A 15 Å vacuum layer was also included in the interface model. The Brillouin zone was sampled using 3 x 3 x 1 *k*-points mesh for geometry optimization.

All the all-atom MD simulations were based on a general AMBER force field[36] with the RESP charges[37] and were carried out using the Gromacs-2023.6 software package[38]. The total run time was 100 ns NVT for the equilibrium MD simulation at 353.15 K. We used the relaxed system as a starting configuration. As it is before system relaxation MD, energy minimization was carried out with a composite protocol of steepest descent using termination gradients. The V-rescale thermostat[39] was used to maintain the equilibrium temperature at 353.15 K. The periodic boundary conditions were imposed on all three dimensions. The Particle Mesh–Ewald method[40,41] was used to compute long-range electrostatics within a relative tolerance of 1×10^{-6}. A cutoff distance of 1.4 nm was applied to real-space Ewald interactions. The same value was used for van der Waals interactions. The LINCS algorithm[42] was applied to constrain bond lengths of hydrogen atoms. A leap-frog algorithm[43] was used with a time step of 2 fs.

Estimation of capture coefficients

Our investigation utilizes a model similar to that described in ref. 19. The parameters employed are detailed in Supplementary Table 1 and largely coincide with those used in ref. 19. This model, originally derived by ref. 20, provides an analytic expression for multiphonon transitions based on a semiclassical approximation. Specifically, it is formulated using Fermi's Golden Rule, assuming a single vibrational mode (the longitudinal optical mode of the perovskite) and incorporating the average thermal occupation of vibrational states. The resulting transition rate, denoted as *r* (per second), is then calculated by

$$r = \frac{|V(Q_c)|^2}{\hbar^2\omega}\sqrt{\frac{2\pi}{py}}\exp\left(p(\theta+\mu-x\cosh(\theta)-\log\left(\frac{1+y}{x}\right)\right), \quad (1)$$

and

$$\theta\frac{\hbar\omega}{k_B T}, \quad (2)$$

$$x = \begin{cases} \frac{S}{p\sinh(\theta)} for\, S < p \\ \frac{p}{S\sinh(\theta)} for\, S > p \end{cases}, \quad (3)$$

$$y = \sqrt{1+x^2} \quad (4)$$

With the phonon energy $E_{ph} = \hbar\omega$ and an energetic difference ΔE between CB/LUMO to trap level or trap level to VB/highest occupied molecular orbital, the expression $p = \Delta E/E_{ph}$ indicates the number of involved phonons for the transition of a charge carrier into the trap. The electronic matrix element $V(Q_c)$ will be estimated by

$$V(Q_c) = \sqrt{\frac{V_T}{V}}\Delta E, \quad (5)$$

where V_T is the volume of the defect. It can be modelled as sphere with a roughly estimated radius $\frac{a^* v}{2}$, with an effective Bohr radius

$$a^* = \frac{\varepsilon_r}{m_{eff}}\frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}, \quad (6)$$

For so-called Huang–Rhys factor *S*, an analytic expression can be found. In case of polar coupling (concerning the interaction of the

monopole caused by a charged defect and the dipole caused by optical phonon modes) a derivation is given in ref. 44.

$$S = S_0 I\left(-2, 2\mu, q_D a^* \frac{v}{2}\right), \tag{7}$$

with

$$S_0 = \frac{3}{2}\frac{1}{(\hbar\omega)^2}\frac{K^2}{M_r\omega/\hbar} \tag{8}$$

with

$$K^2 = e^2\left(\frac{M_r}{V_0}\right)\omega^2\gamma/q_D^2, \tag{9}$$

and

$$I(a,b,c) = \frac{1}{(bc)^2}\int_0^1 \frac{x^a \sin^2\left(b\tan^{-1}(cx)\right)}{\left(1+(cx)^2\right)^b}dx. \tag{10}$$

Here, V_0 is the unit cell volume, $q_D = (6\pi^2)^{1/3}/a_0$ the approximate radius of the Brillouin zone and $v = e/(8\pi\varepsilon a^* \Delta E)^{1/2}$. The so-called Pekar factor $\gamma = \varepsilon_\infty^{-1} - \varepsilon^{-1}$ incorporates the high- and low-frequency limits of the permittivity. The charge of the defect is introduced by $\mu = 0$ for neutral defects and $\mu = \pm v$ if charged.

With the presented model, we estimate the capture coefficients $C = rV$, commonly used for rate equation models, employed in semiconductor-optics software as, for example, SETFOS. The used parameters in Supplementary Table 1 are suited for a perovskite bulk; however, the model cannot take the locally varying permittivity ε at the interface into account. We need to emphasize that our main focus is a qualitative discussion about general trends and the simplicity of the presented model allows us to study the dependence of capture coefficient C on ΔE. Latter energy difference will decrease, when switching from isolated to periodic C_{60}, as shown in Fig. 1a–d.

More extended methods (for example, ref. 45) can incorporate the precise atomic structure of a perovskite/C_{60} interface, including point defects. Such a calculation was carried out in ref. 46, but is barely feasible for our system.

Additional to the results in the main text (Fig. 1h), we provide an additional graph in Supplementary Fig. 10, where the phonon energy was increased from 16.5 meV (solid) to 55.0 meV (dashed), corresponding to Si [44].

Film characterizations

GIWAXS measurements were performed at beamline BL03HB of the Shanghai Synchrotron Radiation Facility using an X-ray wavelength of 0.124 nm recorded with a Pilatus 2M detector. The incident angle was 0.05°, and the exposure time for each GIWAXS data collection was set to 200 s. TOF-SIMS measurements were performed using a focused ion beam TOF-SIMS spectrometer (ION TOF ToF SIMS 5-100). A Bi^+ ion beam was employed as the ion beam to peel the samples. To enhance the clarity of the C18N signal, the C18N-treated C_{60} sample was subjected to a 30-min treatment.

Cross-sectional TEM samples were meticulously prepared using a dual-beam focused ion beam SEM system (FEI Helios NanoLab 450S), which integrates both an electron beam and a gallium ion beam. The Ga^+ ion beam operated at voltages ranging from 1 to 30 kV for precise sample sectioning, while the electron beam was maintained at 2 kV with a beam current of 0.86 pA to minimize beam-induced damage. To protect the C_{60} structure during milling, a 0.5-µm-thick platinum (Pt) layer was first deposited via electron beam deposition, followed by a 2-µm-thick Pt layer using ion beam deposition for enhanced surface protection. Lamella extraction and thinning were performed using a 30 kV Ga^+ ion beam with current ranging from 0.1 to 63 nA, followed by final polishing at 2 kV (23 pA) to reduce amorphous layer formation. High-resolution imaging, including HR-TEM and high-angle annular dark-field scanning TEM (HAADF-STEM), was carried out using Spectra-300 spherical aberration-corrected TEMs operated at 200 kV. These advanced instruments provided sub-angstrom resolution, enabling atomic-scale structural analysis. To mitigate electron beam-induced damage during imaging, HR-TEM data were collected using a Gatan K3 direct-detection electron-counting camera with dose fractionation, and the total electron dose was controlled below 100 e^- $Å^{-2}$. Elemental mapping was performed using a Super-EDS accessory equipped on the Spectra-300 microscope, featuring dual silicon drift windowless detectors for high-sensitivity energy-dispersive X-ray spectroscopy. In addition, EELS was conducted with a beam current of 10 pA using the K3 direct-detection camera to capture the N K-edge, providing complementary compositional information at high spatial resolution. Meanwhile, the 4D-STEM nanobeam diffraction datasets with ~1-nm electron beam size were collected, which achieved significantly higher spatial resolution than conventional SAED. This enabled precise analysis of the crystal structure at the local scale.

PLQY and suns-V_{OC} measurements were performed using the LP20-32 radiative efficiency meter (QYB). The suns-V_{OC} and pseudo-J–V curves were plotted following established procedures as described in previous reports[5]. TRPL measurements were carried out using a PicoQuant FluoTime 300 spectrometer. The TRPL decay data were fitted using a tri-exponential formula: $Y = A_1\exp(-t/\tau_1)+A_2\exp(-t/\tau_2)+A_3\exp(-t/\tau_3)$ (ref. 47). After optimizing the lens position, steady-state PL and lifetime measurements were performed under ambient conditions using a 515.8-nm laser excitation source. Dark current, transient photocurrent, transient photovoltage, Mott–Schottky and EIS were measured by the all-in-one characterization system (PAIOS, Fluxim AG).

Cross-sectional KPFM experiments were conducted using a custom-built system located inside an argon-filled glovebox. To expose the cross-sectional surface, the devices were cleaved from the film side. During the measurements, bias voltages ranging from −1 V to +1 V were applied while scanning the same area. For data analysis, potential profiles were averaged, and the 0 V profile was subtracted to minimize surface-related effects. The internal electric field was obtained by calculating the first derivative of the net potential change induced by the applied biases. By correlating the potential profiles with the topography, the spatial distribution of the electric field within the device was determined. AFM, c-AFM and KPFM measurements were performed using the Park NX20 system. AFM measurements were carried out in non-contact mode. c-AFM measurements were conducted in contact mode. The cantilever tip bias was carefully monitored before and during measurements, and the current amplifier was tuned and optimized before c-AFM data acquisition. KPFM measurements were conducted under electrostatic force microscopy (non-contact) mode. The work function was calibrated before each measurement using a standard gold reference sample. Signal optimization was performed in advance by applying a bias voltage to the samples before measurement.

fs-TA spectra were measured using a Femto/Nano-TA100 system, with an excitation wavelength set at 405 nm. To improve sample optical penetration, the perovskite films were prepared from a 0.75 M precursor solution. Absorbance and transmittance spectra were recorded using an Agilent Cary 7000 spectrophotometer. Before measurement, the lamp was allowed to warm up for 20 min to ensure stability. Raman spectroscopy was acquired using a Raman spectrometer on glass substrates with a Renishaw inVia Qontor instrument and using an argon (514-nm) laser. Electron spin resonance spectroscopy was performed by a JEOL JES-X320 ESR spectrometer.

Electron mobility of pristine and treated C_{60} films was determined using the space-charge-limited current method on an electron-only device with the architecture of ITO/ALD-SnO_2/200 nm C_{60}/ALD-SnO_x/Ag.

The classic Mott–Gurney equation was employed to determine the mobility: $J = 9\mu\varepsilon_0\varepsilon_r V^2/(8d^3)$, where $\varepsilon_r \approx 8.26$ is the average dielectric constant of the blended film[48], ε_0 is the vacuum permittivity, μ is the electron mobility, V is the applied voltage and d is the film thickness.

Device characterizations

J–V measurements were performed using a Keithley 2400 source meter under illumination from a solar simulator (Enlitech, SS-X) with a light intensity of 100 mW cm^{-2}. Our solar simulator was calibrated by certified Si reference cell. The illumination area was defined using shadow masks with apertures of 0.05 cm^2 and 1.02 cm^2. The J–V measurements were performed in an N_2-filled glovebox. EQE measurements were carried out using a Bentham PVE300-IVT system. The LED light intensity was calibrated before each measurement using built-in silicon and germanium photodiodes. EQE values were calculated by dividing the measured photocurrent by the incident photon flux, which was determined using a calibrated silicon photodiode. MPPT measurements were conducted using an MPP Tracking-4B system (Shenzhen Lancheng Technology) under LED-simulated AM1.5G illumination and the obtained curves were normalized. The devices were encapsulated, and measurements were carried out under ambient conditions at 25 °C and ~60% RH.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information.

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Author contributions

Y.H., F.L. and X.Y., supervised the project. T.W., Y.H. and X.Y. conceived the project and designed the experiments. T.W. fabricated the PSCs and prepared the samples for characterization and certification. R.L. performed the DFT calculations and proposed the mechanism of the band-like state formation induced by C_{60} intermolecular coupling. F.L. and R.G. performed the capture coefficient calculations and proposed the mechanism of interfacial recombination assisted by the C_{60}-derived band-like interfacial states. T.W. and J.C. carried out the chemical and photoelectric characterizations of the materials. J.H. and M.L. conducted the TEM characterization. T.W., X.W., P.W., Z.Z., J.L., R.S. and L.Z. jointly optimized the photovoltaic performance of the devices. Z.S. and X. Gao performed the GIWAXS measurements. Z.D. and X. Guo conducted MPPT stability and AFM tests. X.D., Z.X., M.W. and R.J. performed the TOF-SIMS measurements. H.T. and C.X. carried out the cross-sectional KPFM characterizations. Y.W. assisted with the TEM sample preparation. Z.G. contributed to the early-stage DFT studies. Z.J., L.K.L., Q.Z., C.L. and S.L. assisted

with GIWAXS, EIS and PLQY data analysis. T.W., Y.H., X.Y. and F.L. co-wrote the manuscript. All authors reviewed and commented on the manuscript.

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Competing interests

Y.H. is the founder of Singfilm Solar, a company commercializing perovskite photovoltaics. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41566-026-02030-x.

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Solar Cells Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on Nature Research policies, including our data availability policy, see Authors & Referees.

Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	Aperture areas of 0.050 and 1.017 cm2 were used in this work.
Method used to determine the device area	☒ Yes ☐ No	Black metal aperture masks were used during the J-V measurements.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	We provide J-V in both forward and backward direction in Fig. 4c,e.
Voltage scan conditions *For instance: scan direction, speed, dwell times*	☒ Yes ☐ No	See details in Method.
Test environment *For instance: characterization temperature, in air or in glove box*	☒ Yes ☐ No	See details in Method.
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	No preconditioning condition is used.
Stability of the J-V characteristic *Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.*	☒ Yes ☐ No	We provided MPPT data in Fig.4h.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes ☒ No	Negligible hysteresis was observed.
Related experimental data	☒ Yes ☐ No	Reverse and forward J-V scan can be found in Fig. 4c,e.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes ☐ No	We provide EQE measurement in Fig. 4d.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	We compare the integrated Jsc with one from JV scan. The difference between the integrated Jsc from EQE and Jsc from JV scan is within 1.5 % difference, which is within accuracy confidence of the measurements. Jsc from J-V scan is slightly higher than Integrated Jsc from EQE.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	We didn't fabricate tandem solar cell.

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	See details in Method.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Our solar simulator was calibrated by Si reference cell (certificated by Newport).

Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The champion devices were sent to do certification, the certified device spectra mismatch factor (1.023) can be found in Supplementary Fig. 44.

6. Mask/aperture

Size of the mask/aperture used during testing	☒ Yes ☐ No	Optical aperture masks(0.050 and 1.017 cm2) were used.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	No significant variations was observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	We provide certification data in Supplementary Fig. 44.
A copy of any certificate(s) *Provide in Supplementary Information*	☒ Yes ☐ No	We provide certification data in Supplementary Fig. 44.

8. Statistics

Number of solar cells tested	☒ Yes ☐ No	See details in Fig. 4b.
Statistical analysis of the device performance	☒ Yes ☐ No	We provide statistics in Fig.4b.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions *For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature*	☒ Yes ☐ No	We provided stability data and detailed testing condition in Fig.4h and Supplementary Fig. 47.